Dimension-driven parametric reconstruction — does the combo count? Yes, for a real subset

Reconstructing 3D from stated dimensions + views is real engineering, NOT generation. The catch: TMs/RPSTLs are illustrations + scattered specs — not full dimensioned drawings (TDP).

In your corpus (sample): diameter **846** · threads **~600** · torque **1,706** · dim/spec tables **1,301** · clearances **1,807**

BUT — tolerances: 85 · explicit bolt-circles: 2 → full GD&T / complete dimensioning is rare (these aren't engineering drawings).

WHAT IT RELIABLY GROUNDS

- **Standard / parametric parts (big population)**
  NSN/part# → a standard (bolt, nut, washer, fitting, bearing, connector) → exact geometry from the standard's tables.
- **The distinguishing features themselves**
  Thread size, hole/port COUNT, connector/pin type — stated or countable in the figure. This is the look-alike case, nailed.
- **Key stated dimensions**
  Diameters, lengths, clearances, port/thread sizes that the manual does give — enough to size a parametric family member exactly.

WHERE IT BECOMES INFERENCE

- **Complex castings / housings**
  A TM rarely dimensions every surface of an irregular casting → the model would interpolate the un-stated geometry. Guessing.
- **Under-dimensioned / single illustration**
  One exploded view + no dimensions = depth unrecoverable for that shape.
- **Internal / occluded features**
  What isn't drawn or specified can't be reconstructed.

GROUNDED PIPELINE (Tier 2.5 — slots between the 2D viewer and real-CAD/photogrammetry)

1 · Extract parameters
spec tables (parsable) · NSN→standard identity · countable features (holes/ports) · SME confirms a few key dims

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2 · Parametric CAD generator
code-CAD (CadQuery / build123d / OpenSCAD) — exact, offline, per part-FAMILY template instantiated by the params

→

3 · Confidence-labeled 3D (Three.js)
rotate · angle · zoom-to-feature · linked from the part — SPECIFIED features exact; unstated surfaces shown schematic & flagged

THE GROUNDING RULE (what keeps it honest)

Model only what the documentation specifies. A specified feature (thread, hole count, port, stated dimension) is rendered exactly and cited to its source. Anything not specified is shown as a plain schematic placeholder, clearly marked 'representative — not to scale', and never presented as authoritative geometry. So a reconstructed part is trustworthy precisely where it's labeled trustworthy — and the look-alike difference is always in the trustworthy part.

VERDICT — yes, it accounts for something real:

For the large standard-hardware population and for the exact distinguishing features between look-alikes, dimension-driven parametric reconstruction gives accurate, rotatable, cited 3D from the docs alone — no photos, no CAD sourcing. It does NOT give accurate full 3D of arbitrary complex castings from a TM (that still needs a TDP or photogrammetry). Best use: auto-build standard parts + a 'spec-accurate' model that visibly encodes the look-alike difference; fall back to the 2D viewer where the geometry isn't specified.

Proposal only — nothing built. Companion to diagrams 28–29. Dark (R3). Your call on whether to add Tier 2.5.